

Read questions carefully and answer, showing all steps of your solution

Final Answers with no clear solution get ZERO

Work to 5 decimal places in All Questions

Question 1 : Fill in the Blank Questions (with Answers)

1. The _____ error is the difference between the true value and the approximation.
2. The Taylor Series is used to approximate functions using _____ and its derivatives.
3. The Secant Method uses _____ initial approximations to compute the next point.
4. The _____ method requires the derivative of the function to find the next approximation of the root.
5. A necessary condition for applying the Bisection Method is that the function must change _____ over the interval $[a, b]$.

Question 2:

Calculate the true error and relative error for the approximation of $\sqrt{10}$ using the estimated value $\tilde{x} = 3.16$.

Question 3:

Use the Newton-Raphson method to approximate a root of $f(x) = \cos(x) - x$ starting with an initial guess $x_0 = 0.5$. Perform 3 iterations.

Iteration	x_n	$f(x_n)$	$f'(x_n)$	x_{n+1}	$\epsilon_a (\%)$
1					
2					
3					

Question 4:

Solving a System of Linear Equations with LU Decomposition

Given the following system of linear equations:

$$x_1 + 2x_2 + x_3 = 2$$

$$2x_1 + x_2 + 2x_3 = 1$$

$$3x_1 + 2x_2 + x_3 = 3$$

x_1	x_2	x_3

Question 5:

Using Gaussian Elimination, find the values of x and y for the system:

$$2x + y = 5$$

$$x + 3y = 7$$

x	y

Question 6 :

Consider the polynomial $P(x) = x^4 - 3x^3 + 20x^2 - 19x + 13$

- Use Bairstow's Method to find a quadratic factor of the polynomial. Start with an initial guess of $r = -1$ and $s = 1$. Perform 1 iteration.
- Use the resulting quadratic factor to find two of the polynomial's roots.

(Q.2).

2. The true error is difference between the true value and the approximation.

② The Taylor Series is used to approximate functions using polynomials and the derivatives.

3. The Secant method uses two initial approximations to compute the next point.

4. The Newton-Raphson method requires the derivative of the function to find the next approximation of the root.

5. A necessary condition for applying the Bisection method is that the function must change sign over the interval $[a, b]$.

[Q.2].

⇒ soln: $T.V = \sqrt{10} = 3.16228 / A.V = 3.16 = \bar{x}$

relative error

⇒ $\epsilon_t = \frac{T.V - A.V}{T.V} \Rightarrow \epsilon_t = \frac{3.16228 - 3.16}{3.16228}$

$\epsilon_t = 0.00072$

⇒ True error = $T.V - A.V \Rightarrow \text{True error} = 3.16228 - 3.16$

⇒ 0.00228

[Q.3]

soln: $f(x) = \cos(x) - x / x_0 = 0.5$

⇒ $f'(x) = -\sin(x) - 1$

⇒ [2nd iteration]

$\% \epsilon_{a(1)} = \left| \frac{\text{New.V.} - \text{old.V.}}{\text{New.V.}} \right|$
 $\% \epsilon_a = \left| \frac{0.75522 - 0.5}{0.75522} \right| \times 100$
 $\% \epsilon_a = 33.8\%$

⇒ $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} \Rightarrow x_1 = 0.5 - \frac{f(0.5)}{f'(0.5)}$

⇒ $x_1 = 0.5 - \frac{0.37758}{-1.47942} \Rightarrow x_1 = 0.75522$

⇒ [2nd iteration]

⇒ $x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} \Rightarrow x_2 = 0.75522 - \frac{f(0.75522)}{f'(0.75522)}$

⇒ $x_2 = 0.75522 - \frac{-0.02709}{-1.68545} \Rightarrow x_2 = 0.73915$

$$\% \epsilon_a (2) = \left| \frac{\text{New } V - \text{old } V}{\text{New } V} \right| \times 100$$

$$\Rightarrow \% \epsilon_a (2) = \left| \frac{0.73915 - 0.75522}{0.73915} \right| \times 100$$

$$\Rightarrow \% \epsilon_a (2) = 2.17\%$$

$\Rightarrow$ [3rd iteration]

$$X_3 = X_2 - \frac{f(X_2)}{f'(X_2)} \Rightarrow X_3 = \frac{0.73915 - f(0.73915)}{f'(0.73915)}$$

$$\Rightarrow X_3 = 0.73915 - \frac{-0.00011}{-1.67366} \Rightarrow X_3 = 0.73908$$

$$\Rightarrow \% \epsilon_a (3) = \left| \frac{\text{New } V - \text{old } V}{\text{New } V} \right| \times 100 \Rightarrow \% \epsilon_a (3) = \left| \frac{0.73908 - 0.73915}{0.73908} \right| \times 100$$

$$\Rightarrow \% \epsilon_a (3) = 0.0095\%$$

i	X_0	$f(X_0)$	$f'(X_0)$	X_{i+1}	$\% \epsilon_a$
1	0.5	0.37758	-1.47942	0.75522	33.8%
2	0.75522	-0.02709	-1.68545	0.73915	2.17%
3	0.73915	-0.00011	-1.67366	0.73908	0.0095%

⇒ Ex. 4

⇒ solve $x_1 + 2x_2 - x_3 = 2$

$$2x_1 + x_2 - 2x_3 = 1$$

$$3x_1 - 2x_2 + x_3 = 3$$

$$\{A\} \{x\} = \{B\}$$

$$\begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & -2 \\ 3 & -2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 2 \\ 2 & 1 & -2 & 1 \\ 3 & -2 & 1 & 3 \end{array} \right]$$

$-2R_1 + R_2 \rightarrow$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 2 \\ 0 & -3 & 0 & -3 \\ 3 & -2 & 1 & 3 \end{array} \right]$$

$-3R_1 + R_3 \rightarrow$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 2 \\ 0 & -3 & 0 & -3 \\ 0 & -8 & 4 & -3 \end{array} \right]$$

$-8R_2 + 3R_3 \rightarrow$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 2 \\ 0 & -3 & 0 & -3 \\ 0 & 0 & 12 & 15 \end{array} \right]$$

$$\Rightarrow U = \begin{bmatrix} 1 & 2 & -1 \\ 0 & -3 & 0 \\ 0 & 0 & 12 \end{bmatrix}$$

$$\Rightarrow L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & \frac{8}{3} & 1 \end{bmatrix}$$

$$\Rightarrow [L] \cdot [y] = [B]$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & \frac{8}{3} & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

$$\Rightarrow \cancel{y_1 = 2} \quad \Rightarrow y_1 = 2.000000$$

$$\Rightarrow (y_1) \times 2 + y_2 = 1$$

$$\Rightarrow y_2 = [-3.000000]$$

3)

$$\Rightarrow 3y_1 + \frac{8}{3}y_2 + y_3 = 3$$

$$\Rightarrow (3 \times 2) + \left(\frac{8}{3} \times -3\right) + y_3 = 3$$

$$\Rightarrow y_3 = 3 - 6 + 8$$

$$y_3 = 5.000000$$

$$\Rightarrow [U] \cdot [X] = [Y]$$

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & -3 & 0 \\ 0 & 0 & 12 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -3 \\ 5 \end{bmatrix}$$

~~$$\Rightarrow x_3 = \frac{5}{12}$$~~

$$\Rightarrow (x_3) 12 = 5 \Rightarrow x_3 = \frac{5}{12} = 0.41667$$

$$\Rightarrow -3x_2 = -3 \Rightarrow \boxed{x_2 = 1.00000}$$

$$\Rightarrow x_1 + 2x_2 - x_3 = 2$$

$$\Rightarrow x_1 + (2) - 0.41667 = 2$$

$$\Rightarrow \boxed{x_1 = 0.41667}$$

x_1	x_2	x_3
0.41667	1.00000	0.41667

~~Handwritten scribbles and signatures at the bottom of the page.~~

(Q. 5).

⇒ sol:-

$$2x + y = 5$$

$$x + 3y = 7$$

$$\Rightarrow \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$$

$$\Rightarrow \left[\begin{array}{cc|c} 2 & 1 & 5 \\ 1 & 3 & 7 \end{array} \right] \xrightarrow{-2R_2 + R_1}$$

$$\Rightarrow \left[\begin{array}{cc|c} 2 & 1 & 5 \\ 0 & -5 & -9 \end{array} \right] \Rightarrow (0)x_1 + (-5)y_2 = -9$$

$$\Rightarrow y = \frac{-9}{-5} \Rightarrow \boxed{y = 1.8}$$

$$\Rightarrow 2x + y = 5 \Rightarrow 2x + 1.8 = 5$$

$$\Rightarrow 2x = 3.2 \Rightarrow x = \frac{3.2}{2} \Rightarrow \boxed{x = 1.6}$$

x	y
1.6	1.8

Ans

[ex. 6].

⇒ solve

$$P(x) = x^4 - 3x^3 + 20x^2 - 19x + 13$$

$$V_0 = -1 / S_0 = 1$$

	1	-3	20	-19	13
-1		-3	4	-25	48
1		-1	1	-4	25
	1	-4	25	-48	86
-1		-1	5	-31	
1		-1	1	-5	
	1	-5	31	-84	
		1			

$b_0 = 86$
 $b_1 = -48$
 $b_2 = 25$
 $b_3 = -4$
 $b_4 = 1$
 $c_1 = -84$
 $c_2 = 31$
 $c_3 = -5$
 $c_4 = 1$

$$\Rightarrow \begin{cases} c_1 \Delta r + c_2 \Delta s = -b_0 \\ c_2 \Delta r + c_3 \Delta s = -b_1 \end{cases}$$

$$\Rightarrow \begin{cases} -84 \Delta r + 31 \Delta s = -86 \\ 31 \Delta r - 5 \Delta s = 48 \end{cases} \rightarrow \begin{matrix} \text{من الـ 2} \\ \text{النسبة} \end{matrix}$$

$$\begin{bmatrix} \Delta r = 1.95564 \\ \Delta s = 2.5249 \end{bmatrix}$$